

Numerical Simulation Methods And Their Applications

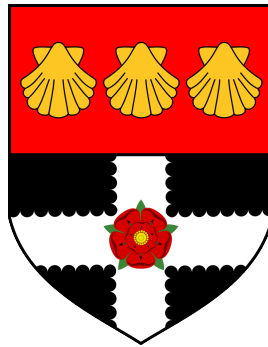
Monte Carlo Methods and Applications in Particle-based Systems

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Plain language summary

This project concerns working with a computer simulation of a particle-based system, particularly using numerical techniques known as Monte Carlo simulations.

It starts with an introduction to the history of these methods, touching upon the infamous Manhattan Project, then focusing on pseudo-random number generators, sampling methodologies, and their applications in statistical mechanics. The project also explores the Metropolis method and its application to a Lennard-Jones liquid. Finally, it concludes with findings of a simulation created from individually developed C++ code.

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1 Development Of Computer Simulation Methods

The first concepts of Monte Carlo simulations can be found in the 18th century, contained within a report found in the 1733 edition of "Histoire de L'Académie Royale des Sciences". This report - created by Georges Louis Leclerc, Comte de Buffon - makes mention of an approximation of pi by throwing needles or rods atop a floor of boards, measuring the fractions that fell on a single board. Of course, this can be recreated in the modern day by a simulation of approximating pi using many points. When initially proposed in this report, the field of Monte Carlo methods had not been distinctly named or partitioned.

1.1 The Manhattan Project

1.1.1 Background

Before understanding Monte Carlo Methods, and the higher level category of "Computer Simulations", we are bound to mention **The Manhattan Project**. The top-secret research and development program was the first to produce Nuclear Weapons, and was a key playing card in the context of the greater geopolitical climate: World War II.

Funded by nations such as the UK, Canada and, majorly, the USA, the Manhattan Project is often cited as a turning point in human history. Disregarding moral arguments concerning atomic weaponry, a large factor in the awe of the project was the pure advancement in scientific knowledge - building upon previous discoveries of fission.

1.1.2 Stanisław Ulam, Von Neumann and Metropolis

1.1.3 Subsequent Developments

2 Monte Carlo Methods

2.1 Pseudo-Random Number Generators

Random number generation is an important part of Monte Carlo methods. There are multiple different methods to obtaining random numbers, but the algorithms we are focusing on within this report are called **Pseudo-Random Number Generators** (PRNGs). While they do not generate "actual" random numbers, what they generate is sufficient enough in variability that it can be used in scenarios where random numbers are needed.

The way that they achieve this is through the use of mathematical formulae, varying in levels of complexity. The algorithms can get increasingly elaborate, yet the simplest can consist of only a few lines of code utilising basic mathematical arithmetic. Sequences are generated which attempt to approximate random numbers, based on an initial starting *seed*.

While humans can not predict the next number in the sequences that are generated, this is not perfectly computationally secure. This is because if the initial *seed state* is known, an identical sequence can be reproduced. Therefore, the sequences are deterministic and not as unpredictable as systems with true chaos built into them - such as radioactive atom decay, or, famously, *Cloudflare's* wall of lava lamps (Liebow-Feeser 2017).

2.1.1 Linear Congruential Generator (LCG)

2.1.2 Mersenne Twister

2.2 Sampling Methodology

2.2.1 Simple

2.2.2 Importance

$$I = \int_a^b f(x) dx$$

2.3 Statistical Mechanics

2.4 The Metropolis Method

3 Question 3? Lennard-Jones ?

4 Practical Application in a Numerical Simulation

4.1 Code and development

4.2 Application to a Lennard-Jones liquid

4.2.1 Numerical Analysis

5 Conclusions, Findings, Prospectives

6 Bibliography

- Frenkel, D. & Smit, B. (1996), *Understanding Molecular Simulation: From Algorithms to Applications*, Academic Press.
- Gentle, J. E. (2003), *Random Number Generation and Monte Carlo Methods*, second edn, Springer.
- Kopka, H. & Daly, P. W. (1999), *A Guide to LaTeX*, third edn, Pearson Education.
- Landau, D. P. & Binder, K. (2021), *A Guide to Monte Carlo Simulations in Statistical Physics*, fifth edn, Cambridge University Press.
- Liebow-Feeser, J. (2017), 'Randomness 101: Lavarand in production', <https://blog.cloudflare.com/randomness-101-lavarand-in-production/>.
- Metcalfe, T. (2023), 'What was the manhattan project?', *Scientific American*.
- Sharma, V. (2023), 'Estimating pi using monte carlo methods', <https://medium.com/the-modern-scientist/estimating-pi-using-monte-carlo-methods-dbd626c888d6>.
- Summerscales, O. (2023), 'Hitting the jackpot: The birth of the monte carlo method', <https://www.lanl.gov/media/publications/actinide-research-quarterly/1123-hitting-the-jackpot-the-birth-of-the-monte-carlo-method>.
- Wang, C. (2018), *Applications of Monte Carlo Methods in Studying Polymer Dynamics*, PhD thesis, University of Reading.